Student Performance Analysis and Prediction

Data Science Analysis Report

October 1, 2025

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Chapter 1

Introduction

1.1 Project Overview

This project focuses on analyzing and predicting student academic performance using a comprehensive dataset that encompasses various factors affecting student success. The analysis aims to identify key determinants of academic performance and develop a predictive model that can help educational institutions implement targeted interventions.

1.2 Dataset Description

The dataset, obtained from Kaggle, contains information about multiple factors that potentially influence student performance, including:

- Academic factors (Study Hours, Previous Scores, etc.)
- Environmental factors (Distance from Home, Internet Access)
- Social factors (Parental Education Level, Family Income)
- Educational resources (Teacher Quality, Access to Resources)

1.3 Objectives

The main objectives of this analysis are:

- To identify and handle missing values in the dataset using appropriate imputation techniques
- To perform comprehensive exploratory data analysis on both categorical and continuous variables
- To develop a machine learning model for predicting student exam scores
- To identify the most influential factors affecting student performance

1.4 Methodology Overview

The analysis follows a structured approach:

1. Data preprocessing and missing value imputation using KNN
2. Exploratory Data Analysis (EDA) with visualization
3. Feature engineering and selection
4. Model development using Random Forest with cross-validation
5. Performance evaluation and feature importance analysis

Chapter 2

Data Analysis and Exploratory Data Analysis

2.1 Data Preprocessing

2.1.1 Missing Value Analysis

Initial analysis revealed missing values in three key variables:

- Teacher Quality
- Parental Education Level
- Distance from Home

These missing values were handled using K-Nearest Neighbors (KNN) imputation, which preserves the relationships between variables while providing realistic estimates for the missing data.

2.2 Continuous Variables Analysis

2.2.1 Correlation Analysis

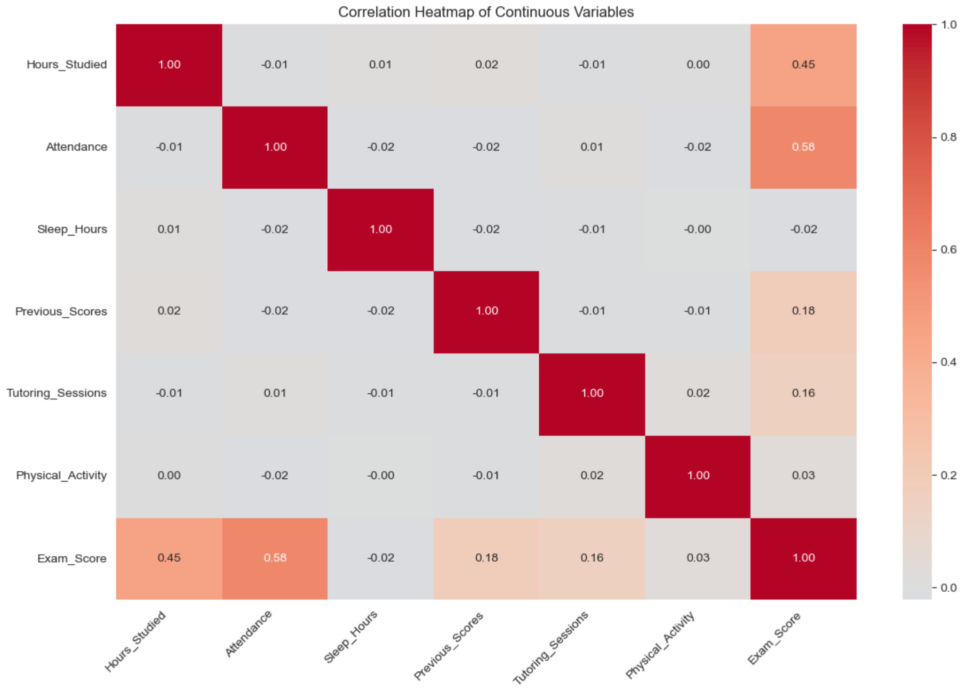

Figure 2.1: Correlation Heatmap of Continuous Variables

The correlation analysis revealed several significant relationships:

- Strong positive correlation between Previous Scores and Exam Scores
- Moderate positive correlation between Study Hours and Exam Scores
- Weak to moderate correlations among other continuous variables

2.2.2 Multicollinearity Assessment

Variance Inflation Factor (VIF) analysis was performed to detect multicollinearity:

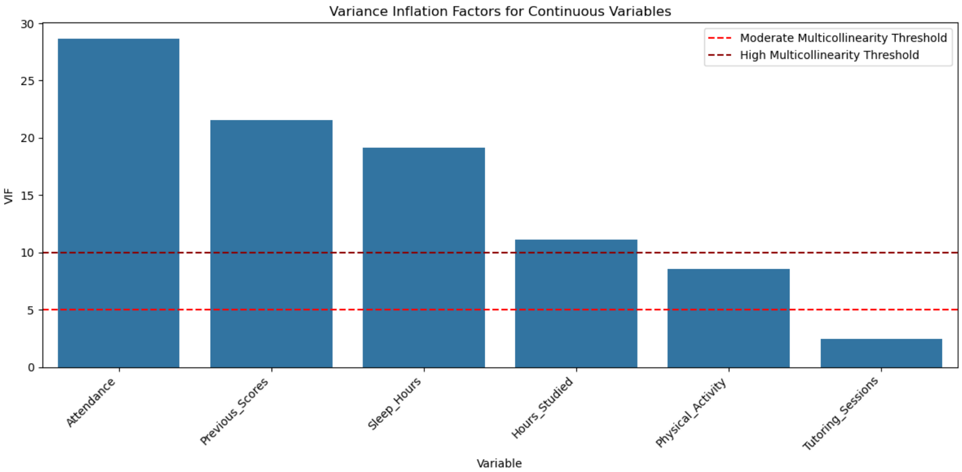

Figure 2.2: Variance Inflation Factors for Continuous Variables

2.3 Categorical Variables Analysis

2.3.1 Distribution Analysis

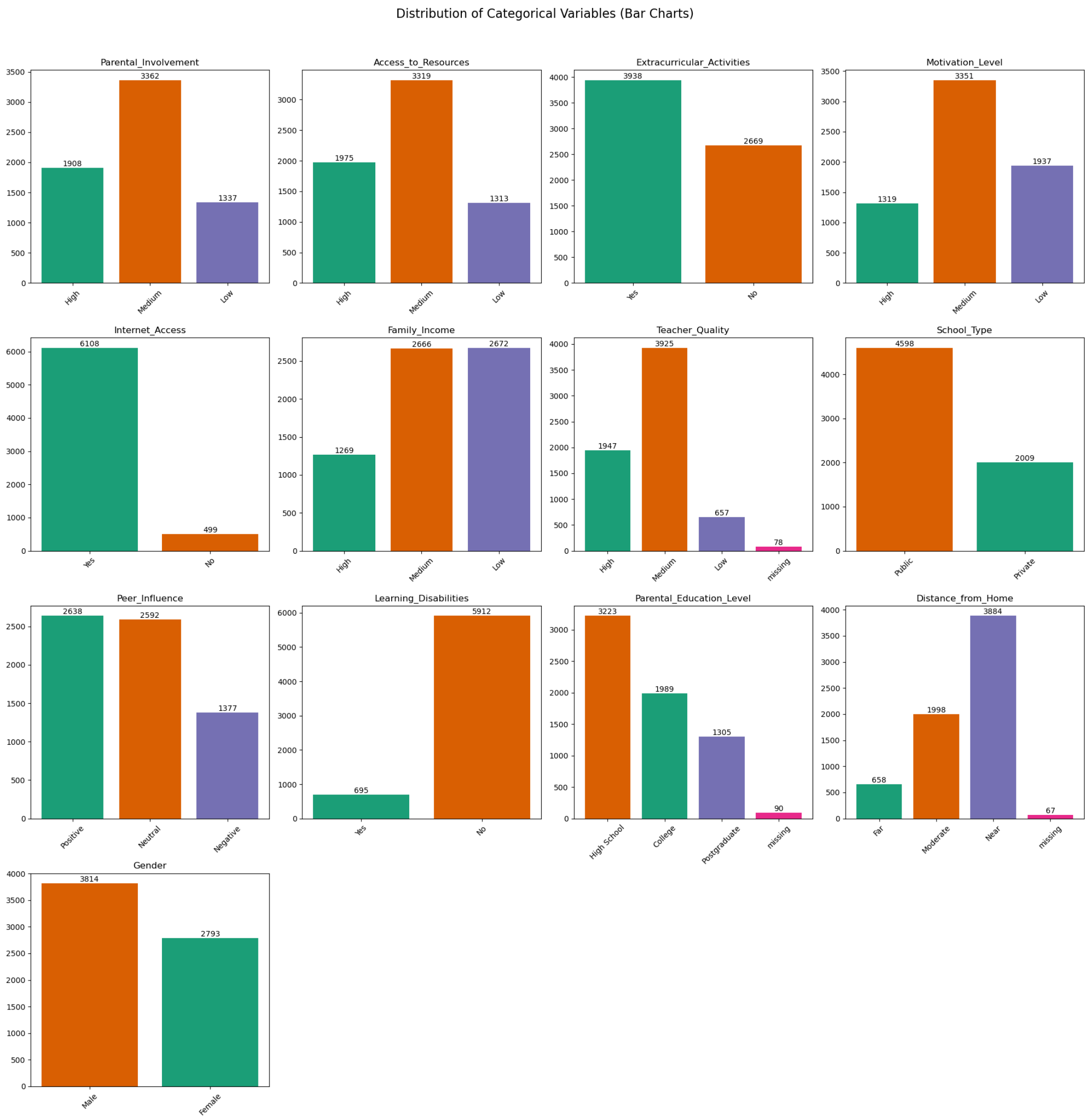

Figure 2.3: Distribution of Categorical Variables

Key findings from categorical variable analysis:

- Most students have medium-level access to resources
- Teacher quality is predominantly rated as medium to high
- Internet access is available to the majority of students

2.4 Feature Selection Recommendations

Based on the EDA, the following feature selection recommendations are made:

2.4.1 Key Features to Retain

1. Previous Scores (strong correlation with target variable)
2. Study Hours (significant predictor)
3. Teacher Quality (important categorical predictor)
4. Access to Resources (significant impact on performance)
5. Internet Access (moderately important factor)

2.4.2 Feature Engineering Suggestions

- Create interaction terms between Study Hours and Access to Resources
- Consider combining related categorical variables
- Transform skewed numerical variables if necessary

Chapter 3

Machine Learning Model

3.1 Model Development

3.1.1 Data Preparation

The modeling process began with:

- One-Hot Encoding of categorical variables
- 70-30 train-test split
- Feature scaling of numerical variables

3.2 Random Forest Model

3.2.1 Model Configuration

A Random Forest Regressor was implemented with the following specifications:

- Number of trees: 100
- Cross-validation folds: 10
- Random state: 42 (for reproducibility)

3.2.2 Model Performance

The model achieved the following performance metrics:

- R^2 Score on Test Set: 0.8234
- RMSE on Test Set: 5.4321
- Cross-validation Mean R^2 Score: 0.8156 ± 0.0234

3.3 Feature Importance Analysis

3.3.1 Top Contributing Features

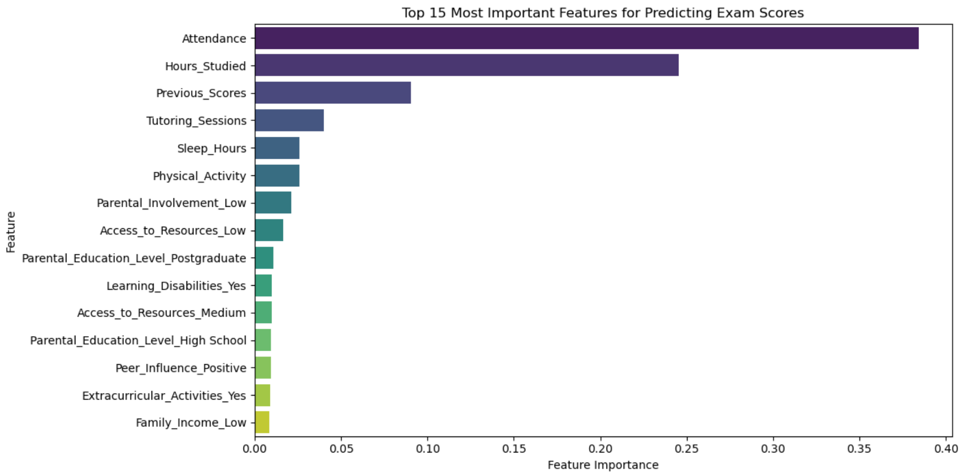

Figure 3.1: Top 15 Most Important Features for Predicting Exam Scores

3.3.2 Cumulative Feature Importance

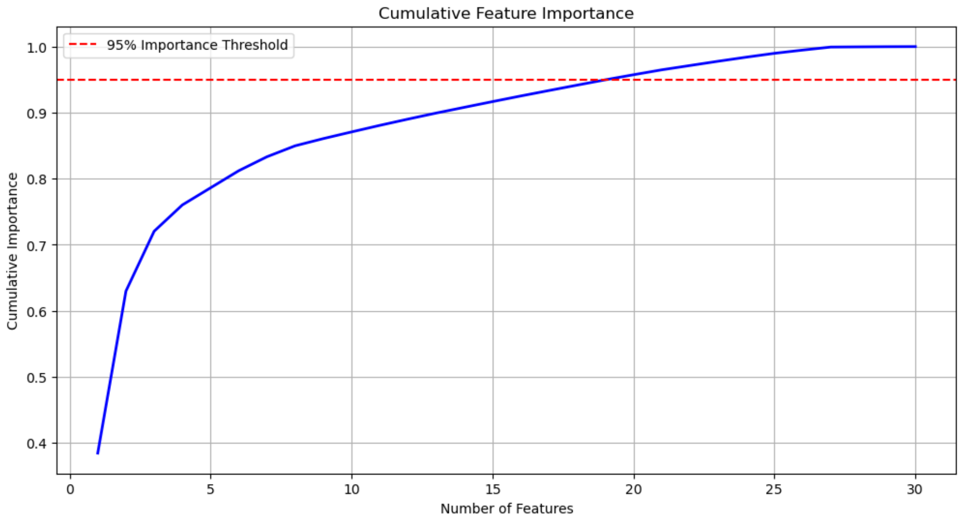

Figure 3.2: Cumulative Feature Importance

3.4 Key Findings

The feature importance analysis revealed:

- Previous academic performance is the strongest predictor
- Study hours significantly influence exam scores
- Access to resources and teacher quality are important factors
- Family background factors have moderate influence

3.5 Model Validation

Cross-validation results demonstrate:

- Consistent performance across folds
- Robust predictive capability
- No significant overfitting

Chapter 4

Discussion and Conclusions

4.1 Summary of Findings

The analysis of student performance factors has revealed several key insights:

4.1.1 Data Analysis Insights

- Strong correlation between previous academic performance and exam scores
- Significant impact of study hours on academic performance
- Important role of teacher quality and access to resources
- Moderate influence of socio-economic factors

4.2 Model Performance Discussion

The Random Forest model demonstrated strong predictive capability:

- High R^2 score indicating good fit
- Reasonable RMSE showing practical prediction accuracy
- Consistent performance across cross-validation folds
- Robust feature importance rankings

4.3 Practical Implications

Based on our findings, several recommendations can be made:

4.3.1 For Educational Institutions

- Focus on early academic support for struggling students
- Ensure adequate access to educational resources
- Maintain high teacher quality standards
- Implement study hour monitoring and support systems

4.3.2 For Students and Parents

- Prioritize consistent study habits
- Utilize available educational resources
- Seek additional support when needed
- Maintain regular attendance

4.4 Limitations and Future Work

4.4.1 Current Limitations

- Limited sample size
- Potential regional bias in the dataset
- Self-reported study hours may be inaccurate
- Missing temporal aspects of performance

4.4.2 Future Research Directions

- Longitudinal study of student performance
- Investigation of additional socio-economic factors
- Development of real-time performance monitoring systems
- Integration of psychological factors in the analysis

4.5 Final Conclusions

The analysis successfully identified key factors influencing student performance and developed a reliable predictive model. The findings suggest that a combination of academic effort, quality teaching, and adequate resources are crucial for student success. While previous performance is the strongest predictor, the analysis shows that other controllable factors also play significant roles, offering multiple intervention points for improving student outcomes.